

COURSE DETAILS

Faculty	: Information and Communication Technology
Programme	: BSc in Software Engineering with Multimedia BSc in Information Technology
Year/ Semester	: Year 4 Semester 2 and Year 3 Semester 2
Lecturer	: Mr. Makheka
Principal Lecturer	: Mr Nthunya
Pre-requisite	: N/A
Co-requisite	: None
Methods of Delivery	: Lectures and Tutorials sessions
Credits	: 12/10
Contacts hours	: 3 (2 hours lecture +1 hour tutorial)
No. of weeks	: 17 weeks long Semester (11 teaching, 1 midterm, 1 Non-Contact, 2 Revision, 2 Final Exams)
No. of Tests	: 1
Assignment	:1
No. of exams	: 2
Portfolio	: None

Prepared by : Mr. Makheka

Checked by Mr Nthunya

Signature :  Date: 30/01/26

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This document comprises the following:

- Essential Information
- Specific Course Information
- Course Rules & Regulations
- Grades
- Plagiarism
- Course Introduction
- Course Aims & Objectives
- Learning Outcome
- Specific Generic Learning Skills
- Syllabus + Lecture Outline
- References
- Assignment Schedule
- Assessment Criteria
- Specific Criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for Assignments
- Submission Requirements + Guidelines

1.0 ESSENTIAL INFORMATION

- All courses other than electives are '**significant courses**'
- As an indicator of workload one credit carries an additional 2 hours of self study per week. For example, a course worth 10/12 credits require that the student spends an additional 20/24 hours per week, either reading, completing the assignment or doing self directed research for that course.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- **Re-submission** of work can only receive a **50% maximum pass rate**.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.

2.0 SPECIFIC COURSE INFORMATION

- Attendance rate of 80% is mandatory for passing course.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 COURSE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the UNIT AIMS AND OBJECTIVES
- All tests/examinations are compulsory.
- Students must sit for the test/examination on the notified date.
- Students are expected to familiarize themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is **TOTALLY** the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials **WITHOUT** appointments will also **NOT** be entertained.

4.0 GRADES

All modules and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PA or F:

<i>Grade</i>	<i>Numeric Grade</i>	<i>Description</i>
90 – 100	A+	Pass with Distinction
85 – 89	A	
80 – 84	A-	
75 – 79	B+	Pass with Credit
74 – 70	B	
65 – 69	B-	
60 – 64	C+	Pass
55 – 59	C	
50 – 54	C-	
40 – 47	PP, X	Fail
0 – 39	F	

EXP Exempted
 PP Pass Provisional with extra work needed
 PX Pass after extra work is given and passed
 X Extra work should be given
 D Deferred
 W Withdraw
 DNA Did Not Attend Course
 DNC Did Not Complete Course
 ANN Results Annulled due to misconduct

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

5.0.1 Academic Dishonesty

Academic Dishonesty or academic misconduct is any type of cheating that occurs in relation to a formal academic exercise. It can include:

Plagiarism: The adoption or reproduction of original creations of another author (person, collective, organization, community or other type of author, including anonymous authors) without due acknowledgement.

Fabrication: The falsification of data, information or citations in any formal academic exercise.

Deception: Providing false information to an instructor concerning a formal academic exercise – e.g. giving a false excuse for missing a deadline or falsely claiming to have submitted work

Cheating: Any attempt to give obtain assistance in a formal academic exercise (like an examination) without due acknowledgement.

Bribery: or paid services. Giving certain test answers for money.

Sabotage: Acting to prevent others from completing their work. This includes cutting pages out of library books or wilfully disrupting the experiments of others.

Students involved in academic dishonesty will be treated as follows:

1. Plagiarism and fabrication in Assignments, Projects or Presentations: A zero (0) mark will be awarded to the work.
2. Deception: 10 % will be deducted on the first day of late submission and 5% a day thereafter. Late submission on or after BOS will not be accepted.
3. Cheating: Any form of assistance in a test will be rewarded a zero (0) mark. While in an examination the student automatically repeats the module.
4. Bribery, Sabotage or Professional Misconduct: Committing any of these offenses will lead to discontinuity from the University for a certain period depending on the seriousness of the offence.

5.0.2 Copyright, Patents and Ownership of Work

All use of the borrowed work, whether text or images must be referenced and acknowledged. Limkokwing is not liable for any infringement of copyright within students' work. Students are to abide by International Copyright laws. Students engaged in a project activity will be required, as a condition of acceptance as a student, to agree to assign to Limkokwing their right, title and interest in any invention or visual work arising from their studies with the University. Limkokwing has the right to keep students work for use in archives, exhibitions and events. Where students wish to keep original work they must apply in writing and gain permission from the Academic Management.

5.0.3 PROGRESSION POLICY

Students can only enroll into a higher semester if they have passed all courses in the preceding semester or have failed not more than two courses. A student cannot register for any course at a

Course Outline

BIAI4210/BIAI3210-Artificial Intelligence

higher level if they fail its lower level unit which in essence is a pre-requisite. If student CGPA is less than 2.0 could not proceed to the next semester.

6.0 COURSE INTRODUCTION

This course introduces the fundamental concepts, principles, and techniques that have shaped the development of Artificial Intelligence. It begins by examining the historical foundations of AI, key definitions, and the major dimensions of artificial intelligence, highlighting how AI differs from traditional programming approaches and general computational paradigms. Students are introduced to production systems as a core mechanism for modelling intelligent behaviour in AI systems.

The course further explores knowledge representation and reasoning, focusing on resolution as a fundamental rule of inference for well-formed logical formulas. It also introduces planning as an abstract problem-solving approach used to model and reason about goal-directed behaviour. Finally, the course provides an introduction to machine learning, addressing the fundamental question of how computer systems can automatically learn and improve their performance through experience.

7.0 COURSE AIMS AND OBJECTIVES

- Understand the basics of artificial intelligence and its subfields.
- Equip students with problem-solving techniques based on search, reasoning, and planning
- Introduce machine learning concepts as a core component of modern AI systems
- Promote awareness of the ethical, social, and professional implications of AI technologies
- Prepare students for advanced AI, machine learning, and applied AI courses
- Understanding of these AI techniques: Expert systems, Neural Network, Evolution computing, Virtual Reality and Virtual Engineering, Cognitive Science, Data Modeling, Fuzzy Logic and Control and Emerging Computing using Agent.
- Introduce Logic Programming languages, in particular Prolog

8.0 LEARNING OUTCOMES

Upon completion of the Course, student will:

1. Explain the core concepts, terminology, and historical development of Artificial Intelligence
2. Model problems as state-space representations and apply uninformed and informed search algorithms
3. Design and analyze heuristic functions for efficient problem solving
4. Represent knowledge using logical and rule-based formalisms and perform basic automated reasoning
5. Design and evaluate simple intelligent agents for problem-solving tasks
6. Apply introductory machine learning techniques and assess their performance using appropriate metrics
7. Analyze ethical, social, and professional issues related to the development and deployment of AI systems

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of the Course, student will acquire skills in:

- Knowing the history of AI.

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- Knowing the definition of AI.
- To liaise in a few of these AI topics: Expert systems, Neural Network, Evolution computing, Virtual Reality and Virtual Engineering, Cognitive Science, Data Modeling, Fuzzy Logic and Control and Emerging Computing using Agent.
- Allow for further developing knowledge to solve problems using AI techniques if continuous study is pursue

10.0 UNIT SYLLABUS + LECTURE OUTLINE:

Week:	1
Lecture 1:	AN INSIGHT INTO THE COURSE
Lecture Synopsis:	Definition and scope of AI Historical overview and key milestones Differentiating AI from human intelligence AI Agents Characteristics of Intelligent Agents AI Techniques and Applications - Fuzzy Logic & Fuzzy Systems - Fuzzy system applications - Virtual Reality and Virtual Engineering - Computer Vision Applications of AI and Ethics of AI AI in healthcare: Diagnosis, treatment, and medical imaging AI in finance: Fraud detection, algorithmic trading, and risk assessment AI in transportation: Autonomous vehicles and traffic optimization Ethical concerns in AI development Societal impacts of AI
Recommended Reading(s):	✓ Chapter(s) – 1; Artificial Intelligence: A Modern Approach ✓ Chapter(s) – 2; Artificial Intelligence: A Modern Approach
Tutorials	None
Case Study(s)	Introducing the tools to be used
Week:	2
Lecture 2:	EXPERT SYSTEMS
Lecture Synopsis:	Definition of Expert System Rule-based Expert system - Features - Inference techniques - Implementations Numerical based Expert System - Features - Implementations
Recommended Reading(s):	✓ Chapter(s) 1 to 4; Expert Systems: Principles and Programming ✓ Chapter(s) 1 to 2; Programming with Prolog ✓ Chapter(s) 1; Learn Prolog Now!
Week:	3
LECTURE 3:	KNOWLEDGE REPRESENTATION
Lecture Synopsis:	• Need for Knowledge Representation • Approaches: ▪ Frames ▪ Semantic networks ▪ Production rules • Introduction to Logic-based representation:

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	- Propositional logic - Predicate logic (FOL) - Prolog - Fuzzy Reasoning
Recommended Reading(s):	✓ Chapter(s) – 7 to 10; Artificial Intelligence: A Modern Approach
Week:	4-5
Lecture 4 - 5:	PLANNING AND SEARCH METHODS
Lecture Synopsis:	Problem Spaces and Searching - Search Problem Formulation - Uninformed Search Strategies: - Breadth-First Search - Depth-First Search - Uniform Cost Search - Informed Search Strategies: - Best-First Search - Greedy Search - A* Algorithm - Heuristics and Admissibility - Local Search Algorithms: - Hill Climbing - Simulated Annealing - Game AI - Minimax algorithm - Alpha beta pruning
Recommended Reading(s):	✓ Chapter(s) – 4; Artificial Intelligence: A Modern Approach ✓ Chapter(s) – 5; Artificial Intelligence: A Modern Approach
Week:	6
Lecture 6:	ROBOTICS
Lecture Synopsis:	Definition of Robot and Robotics Robot Hardware and Control Systems - Microcontrollers - Actuators and Motors - Sensors for Perception
Lab Activity:	STEM Lab
Week:	7
	MIDTERM EXAMINATION
Week:	8
	SEMESTER BREAK
Week:	9
Lecture 7:	DATA ANALYTICS
Lecture Synopsis:	Data Mining - Data Preprocessing - Data Cleaning - Data Integration - Data Reduction - Data Transformation and Discretization Data Visualization Techniques - Pixel-Oriented Visualization - Icon Based Visualization

Course Outline

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	Mining Frequent Patterns, Association and Correlations
Recommended Reading(s):	Data Mining Concepts and Techniques Chapter 2 ,3 and 6
Tutorials	Lab 1
Week:	10
Lecture 8:	MACHINE LEARNING
Lecture Synopsis:	• Introduction to machine learning - What is ML? Types: Supervised, Unsupervised, Reinforcement • ML Pipeline: data, features, models, evaluation • Use cases: Recommendation, Spam filtering, etc
Recommended Reading(s):	Introduction to Machine Learning with Python
Week	11
Lecture 8	MACHINE LEARNING – SUPERVISED LEARNING
Lecture Synopsis:	• Linear Regression • Logistic Regression • Model evaluation: accuracy, precision, recall, F1-score • Overfitting, underfitting, cross-validation
Recommended Reading(s):	Introduction to Machine Learning with Python
Week	12
Lecture 8	MACHINE LEARNING - UNSUPERVISED LEARNING
Lecture Synopsis:	• Clustering: K-Means, DBSCAN • PCA for dimensionality reduction • NLP Basics: ▪ Text preprocessing: Tokenization, Stopword removal, Lemmatization ▪ Vectorization: Bag of Words, TF-IDF
Recommended Reading(s):	✓ Introduction to Machine Learning with Python
Week	13 & 14
LECTURE 13	DEEP LEARNING
	• What is Deep Learning? • Difference between ML and Deep Learning • Artificial Neural Networks (ANN): • Perceptron & Multi-layer Perceptron (MLP) • Architecture: Input → Hidden Layers → Output • Activation Functions: ReLU, Sigmoid, Softmax • Convolutional Neural Networks (CNNs) – Basic idea • Feature extraction, convolutional filters, pooling layers • Use case: Image classification • Recurrent Neural Networks (RNNs) – Basic idea • Sequential data, LSTM overview • Use case: Sentiment analysis
Recommended Reading(s):	✓ Chapter(s) – 21; Artificial Intelligence: A Modern Approach ✓ Introduction to Machine Learning with Python
Week:	15
Lecture 9:	REVISION WEEK
Week:	16
	FINAL EXAMINATION WEEK

Week:	17
	FINAL EXAMINATION WEEK

11.0 REFERENCES

11.0.1 RECOMMENDED BOOKS

1. S. Russell & P. Norvig. Artificial Intelligence: 4th Edition A Modern Approach, Prentice Hall, 2020.
2. Data Mining: Concepts and Techniques, 3rd Edition. Jiawei Han, Micheline Kamber, Jian Pei.

11.0.2 ADDITIONAL BOOKS

1. Michael negnevitsky Artificial Intelligence A Guide to Intelligent Systems Second Edition
2. Data Analysis From Scratch With Python_ Beginner Guide using Python, Pandas, NumPy, Scikit-Learn, IPython, TensorFlow and Matplotlib
3. Patrick Blackburn, Johan Bos, Kristina Striegnitz, Learn Prolog Now, 2001

12.0 ASSESSMENT SCHEDULE

Assessment Description	Issue Date	Due Date	%
Class Test	Week 5	Week 5	10%
Mid Term Examination	Week 7	Week 7	20%
Group Assignment	Week 5	Week 12	30%
Final Examination	Week 16 & 17	Week 16 & 17	40%
Total			100%

13.0 ASSESSMENT CRITERIA

You will be graded based on your effort in submitting successfully well worked assessments.

14.0 SPECIFIC CRITERIA

- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.

